



Effect of 25-OH-cholecalciferol on mineral metabolism of Holstein cows during the transition period

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ABSTRACT

High-yielding dairy cows around parturition are challenged by varying degrees of low calcium status (hypocalcemia), which is associated with increased risk of several metabolic disorders and diseases in early lactation. The aim of the present study was to investigate effects of dietary 25-hydroxycholecalciferol (25-OH-D₃; calcidiol) supplementation during the transition period. Forty-two multiparous Holstein cows were fed 1 of 2 treatments: either cholecalciferol (D₃) or 25-OH-D₃ (HyD) as dietary vitamin D supplements. From 21 d before the expected calving until parturition, the D₃ group received 0.475 mg/d (0.035 mg/kg of DM) of D₃, and the HyD group received 1.45 mg/d (0.108 mg/kg of DM) of 25-OH-D₃. From parturition until 21 DIM, the D₃ group received 0.625 mg/d (0.024 mg/kg of DM) of D₃, and the HyD group received 0.625 mg/d (0.024 mg/kg of DM) of D₃ plus 1.0 mg/d of 25-OH-D₃ (0.056 mg/kg of DM). Arterial blood samples were collected by puncture of *a. auricularis* on d -21, -7, -3, 0 (calving day), and +1 relative to calving for analysis of blood ionized Ca. Blood samples were taken from the tail on d -21 (22 ± 5 d), -7 (6 ± 2 d), -3 (2 ± 1 d), 0 (d 0; 13 ± 8 h after calving), 1 (36 ± 11 h after calving), 10 (9 ± 2 d), and 20 (16 ± 2) relative to calving to analyze plasma 25-OH-D₃ level, plasma minerals, and other plasma metabolites. During the entire experiment, cows were fed TMR, and DMI, productive performance, metabolic disorders, and other diseases were recorded. Milk yield, milk protein percentage, milk lactose percentage, BCS, and BW were not affected by treatment. There was an interaction between treatment and time for DMI. The DMI in HyD was numerically lower compared with D₃ at d -3, -2, and -1 relative to calving; however, the DMI in HyD was numerically higher compared with D₃ at d 0, 1, and 2 postpartum. Plasma 25-OH-D₃ was higher in HyD compared with D₃

both pre- and postpartum; however, plasma 1,25(OH)₂D₃ was not affected by treatment. Arterial blood ionized Ca was affected by treatment, and it was higher in the HyD (1.08 ± 0.02) compared with the D₃ on the day of calving (0.99 ± 0.02 mmol/L; lsmeans ± SEM). There was a tendency for lower incidence of mastitis in HyD compared with D₃. Plasma BHB level (lsmeans ± SEM) was significantly lower in HyD (0.61 ± 0.02) compared with D₃ (0.70 ± 0.02 mM). In conclusion, feeding 25-OH-D₃ to transition dairy cows increased arterial blood ionized Ca on the day of calving compared with cows supplemented with D₃. The increased arterial blood ionized Ca in HyD was apparently mediated by 25-OH-D₃ without affecting plasma 1,25-(OH)₂D₃. Additionally, the results showed a lower incidence of mastitis in HyD.

Key words: transition cow, calcidiol, calcitriol, hypocalcemia, milk fever

INTRODUCTION

High-yielding dairy cows are unable to maintain blood Ca and P concentrations at calving, and the low Ca status (hypocalcemia) remains as one of the major problems, which affects production and health of high-yielding cows. Milk fever (parturient paresis) is a well-known consequence of low Ca status after or close to calving (DeGaris and Lean, 2008). Cows with hypocalcemia are also at higher risk of clinical diseases such as mastitis, retained placenta, displaced abomasum, and uterine prolapse (Thilising-Hansen et al., 2002). Providing a low Ca diet during the close-up period, negative DCAD, and supplementation with vitamin D in dairy cows have been practices to improve the Ca status and reduce the prevalence of milk fever at calving time (DeGaris and Lean, 2008). There have been considerable attempts to prevent hypocalcemia in postpartum dairy cows following the discovery of vitamin D's role in Ca homeostasis, as reviewed by Wilkens et al. (2020).

Vitamin D₃ (cholecalciferol) is the most commonly used vitamin D supplementation in dairy cattle diets. However, after intestinal absorption, vitamin D₃ should undergo 2

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The list of standard abbreviations for JDS is available at adsa.org/jds-abbreviations-26. Nonstandard abbreviations are available in the Notes.

steps of hydroxylation to form its active form (Nelson et al., 2016). The first step of vitamin D₃ hydroxylation takes place in the liver by the 25-hydroxylase enzyme to generate the 25-OH-D₃ (calcidiol). In the second step of vitamin D₃ activation, 25-OH-D₃ undergoes another hydroxylation step in the kidney by 1 α -hydroxylase enzyme to generate the 1,25-(OH)₂-D₃ (calcitriol). The 25-OH-D₃ is the most stable and abundant metabolite from vitamin D₃, with a half-life of 15 d, hence, its suitability as an indicator of vitamin D₃ status of the animal (Nelson et al., 2016). The 1,25-(OH)₂-D₃ acts as a steroid hormone and is the most active form of vitamin D₃. However, conversion of 25-OH-D₃ to 1,25-(OH)₂-D₃ is a tightly regulated process and controlled by parathyroid hormones in response to Ca status (Fraser and Kodicek, 1973).

Plasma levels of vitamin D metabolites are responsive to dietary supplementation of vitamin D₃. Rodney et al. (2018) found that even when D₃ was supplied at ~4 times the NRC (2001) recommendations, plasma 25-OH-D₃ levels did not exceed 100 ng/mL, indicating a regulatory mechanism in the liver's conversion of D₃ to 25-OH-D₃ or conversion of 25-OH-D₃ to 24,25-(OH)₂-D₃ (Rodney et al., 2018; Venjakob et al., 2024). Increasing serum 25-OH-D₃ is more effectively achieved by supplementing with 25-OH-D₃ compared with D₃ (Rodney et al., 2018; Poindexter et al., 2020). Therefore, supplementing with 25-OH-D₃ might be a more potent supplement and may induce additional effects of vitamin D supplementation compared with supplementation of D₃ alone. Achieving adequate vitamin D status, as indicated by plasma 25-OH-D₃ levels (Nelson et al., 2016), is more easily achievable with 25-OH-D₃ supplementation than with vitamin D₃ (Poindexter et al., 2020, 2023a). In conjunction, Wilkens et al. (2012) demonstrated an increase in prepartum serum ionized Ca in cows fed 3 mg of 25-OH-D₃ compared with the control. Similarly, Poindexter et al. (2020) reported that supplementation of 1 or 3 mg of 25-OH-D₃ compared with 1 or 3 mg of D₃ prepartum increased Ca status postpartum. Martinez et al. (2018a) demonstrated a lower incidence of transition disorders and diseases (i.e., retained placenta and metritis) in cows fed 3 mg/d of 25-OH-D₃ compared with 3 mg/d D₃ from 252 d of gestation until calving. This lower incidence of metabolic disorder could be explained by upregulating genes involved in immune response (Vieira-Neto et al., 2021).

In the present study, it was hypothesized that 25-OH-D₃ supplementation during the close-up period and early lactation would increase the plasma 25-OH-D₃ and overcome the rate-limiting hydroxylation of cholecalciferol to 25-OH-D₃ in the liver, and finally the increased plasma 25-OH-D₃ will improve the blood ionized Ca and mineral status of the cows close to calving time. Thus, the objective of the present study was to investigate the effect of 25-OH-D₃ supplements from 3 wk before calv-

ing to 21 DIM on arterial blood ionized Ca, 25-OH-D₃, 1,25(OH)₂D₃, and plasma mineral status.

MATERIALS AND METHODS

Animals and Experimental Design

Forty-two multiparous Danish Holstein cows from the Danish Cattle Research Center, Foulum, Denmark, were used in a completely randomized design. The cows were housed in a loose-housing system with deep straw bedding and a concrete floor and cubicles during the close-up and lactating periods, respectively. At ~2 d before the expected calving date, cows were moved to individual calving pens with straw bedding. All cows were fed the same far-off diet, which was composed (g/kg diet DM) of 912 of grass clover silage, 80 of barley straw, and 8.0 of mineral and vitamin supplement. From 21 d before expected calving date, cows were randomly assigned to 1 of 2 treatments: cholecalciferol (D₃) or supplemented with 25-OH-D₃ (**HyD**). During the close-up period, D₃ group received 0.475 mg/cow/d (0.035 mg/kg DM) of D₃ supplements via vitamin and mineral supplements. In the HyD close-up ration, the mineral and vitamin supplements did not contain D₃, and 25-OH-D₃ was supplied in a dose of 1.45 mg/cow/d (0.108 mg/kg DM), as per EU regulations limiting the application of combined D₃ and 25-OH-D₃ (Bampidis et al., 2023). After calving, D₃ received the recommended vitamin D level (0.625 mg/d/cow equal to 0.024 mg/kg DM), and in HyD group, D₃ diet was supplemented with 25-OH-D₃ (1.0 mg/cow/d equal to 0.056 mg/kg DM). Therefore, during the lactation period, HyD received 0.625 mg/d/cow of cholecalciferol plus 1.0 mg/cow/d of 25-OH-D₃ (0.080 mg/kg DM of D₃ + 25-OH-D₃). All cows were fed the same close-up and lactating TMR diets (Table 1) composed to fulfill the Nordic recommendation for nutrient allowances (Volden, 2011).

Experimental Sampling and Data Collection

All cows were provided with ad libitum access to feed (7%–10% orts), and the daily meals were divided into 2, fed at 0800 and 1600 h. The cows were provided unrestricted access to water. Daily feed intake was recorded throughout the experiment using Insentec feed troughs (Insentec, Marknesse, the Netherlands). A weekly sample of TMR was taken to measure the DM content, and daily DMI was calculated by multiplying feed intake by DM content. Daily DMI was averaged weekly, and it was used in the statistical analysis (3 wk of close-up and 3 wk of lactation). Milk yield was recorded daily from 2 DIM, and weekly milk yield was used for data analysis. A weekly milk (an afternoon and a morning) sample was analyzed for fat percentage, protein percentage, lactose

Table 1. Total mixed ration, analyzed mineral, and calculated nutrient composition of diets

Ingredient (g/kg DM unless noted)	Close-up diet ¹		Lactating diet ²	
	D ₃	HyD	D ₃	HyD
Grass silage	—	—	179.7	179.7
Corn silage	741.1	741.1	344.2	344.2
Barley	—	—	153.0	153.0
Corn, finely ground	74.9	74.9	—	—
Rapeseed cake, 10.5% fat	104.8	104.8	130.0	130.0
Rapeseed meal, 4% fat	—	—	84.1	84.1
Corn gluten	59.9	59.9	—	—
Sugar beet pellets	—	—	91.8	91.8
Salt	—	—	1.8	1.8
Sodium bicarbonate	—	—	9.9	9.9
Ca carbonate	1.9	1.9	0.7	0.7
Mg chloride hexahydrate	4.8	4.8	—	—
Ca chloride dihydrate	5.2	5.2	—	—
Mineral and vitamin premixed ³	7.5	7.5	3.7	3.7
Vitamin A, E, and D supplements	—	—	1.1	1.1
Mineral analysis ⁴				
DM, g/kg	363 ± 15	364 ± 12	395 ± 5	392 ± 11
Ca, g/kg DM	4.6 ± 0.4	4.6 ± 0.3	6.8 ± 1.3	7.3 ± 0.3
P, g/kg DM	2.7 ± 0.2	2.8 ± 0.1	4.85 ± 0.4	4.75 ± 0.2
Mg, g/kg DM	2.2 ± 0.2	2.3 ± 0.2	3.0 ± 0.4	3.1 ± 0.1
K, g/kg DM	8.4 ± 0.8	8.6 ± 0.4	12.5 ± 0.7	12.5 ± 0.7
Na, g/kg DM	2.5 ± 0.2	2.5 ± 0.2	4.45 ± 1.9	4.85 ± 1.1
Total S, mg/kg DM	2.0 ± 0.2	2.0 ± 0.1	2.9 ± 0.2	2.9 ± 0.1
Fe, mg/kg DM	262 ± 29	308 ± 36	425 ± 191	435 ± 35
Mn, mg/kg DM	40.6 ± 3.9	41.4 ± 3.8	59 ± 12.7	62.5 ± 3.5
Zn, mg/kg DM	95.8 ± 17.8	100.1 ± 15.0	53.5 ± 6.4	74 ± 12.7
Cu, mg/kg DM	14.5 ± 2.6	14.2 ± 2.8	6.6 ± 1.5	7.6 ± 0.8
Cl, mg/kg DM	9.6 ± 1.1	9.6 ± 1.2	4.7 ± 0.8	5.0 ± 0.4
Calculated DCAD, mEq/kg DM	-73	-65	+ 202	+ 213
Nutrient composition, calculated				
Energy, MJ/kg DM	7.31	7.31	6.56	6.56
CP, g/kg DM	149.9	149.9	163	163
Protein balance in the rumen, g/kg DM	-2	-2	12	12
Fatty acids, g/kg DM	27	27	26	26
NDF, g/kg DM	310	310	320	320
Starch, g/kg DM	262	262	193	193
Vitamin D ₃ , mg/kg DM	0.035	—	0.024	0.024
25-OH-D ₃ , mg/kg DM	—	0.108	—	0.056

¹The D₃ contained 0.475 mg/cow per day of cholecalciferol. However, in HyD, mineral and vitamin supplements did not contain any cholecalciferol, and 25-OH-D₃ was supplied in a dose of 1.45 mg/cow per day.

²The D₃ contained 0.625 mg/cow per day of cholecalciferol. However, in HyD, mineral and vitamin supplements contained 0.625 mg/cow per day of cholecalciferol, and 25-OH-D₃ was added in a dose of 1.0 mg/cow per day.

³25-OH-D₃ was mixed in the mineral and vitamin premix.

⁴In the close-up and lactating diets, 9 and 2 samples per treatment were analyzed (mean ± SD), respectively. Composition of mineral and vitamin premixed for close-up diet (per kg of DM): NaCl 635 g, heat-treated wheat meal 150 g, sugar beet molasses 30 g, vitamin A 416,700 IU, vitamin D₃ 208,300 IU, synthetic α -tocopheryl acetate 50,000 IU, copper (II) sulfate 2.083 mg, manganese oxide 2.083 mg, zinc oxide 8.333 mg, calcium iodate 208 mg, sodium selenate 50 mg, and cobalt (II) carbonate 58 mg. Composition of mineral and vitamin premixed for lactating diet (per kg DM): calcium carbonate 369 g, NaCl 301 g, magnesium oxide 257 g, calcium magnesium carbonate 8 g, sugar beet molasses 3 g, calcium magnesium carbonate 5 g, vitamin A 600,000 IU, vitamin D₃ 190,000 IU, synthetic α -tocopheryl acetate 4,000 IU, manganese (II) oxide 4,000 mg, copper (II) sulfate 1,500 mg, zinc oxide 4,500 mg, calcium iodate 225 mg, coated cobalt (II) carbonate 25 mg, sodium selenate 50 mg.

(monohydrate) percentage, and SCC at a commercial laboratory (Eurofins Milk Testing Denmark A/S, Galten, Denmark). The ECM was calculated based on the milk composition and milk yield on the day that milk composition was measured. Body weight of cows was recorded twice a day during lactation period, and weekly average of BW was used for data analysis. Body condition score (scale 1–5) was recorded on 21 DIM and used for data analysis (Wildman et al., 1982).

Sampling days of blood were designated to d -21 (22 ± 5 d), -7 (6 ± 2 d), -3 (2 ± 1 d), 0 (d 0; 13 ± 8 h after calving), 1 (36 ± 11 h after calving), 10 (9 ± 2 d), and 20 (16 ± 2) relative to calving date. A tail blood sample was taken on d -21 (n = 40), -7 (n = 40), -3 (n = 39), 0 (n = 40), 1 (n = 40), 10 (n = 40), and 20 (n = 40) relative to calving date, placed in the crushed ice, and centrifuged at 3,000 × g for 15 min at a temperature of 4°C. Afterward, the plasma was harvested. Plasma total Ca, P, and 25-OH-D₃

were analyzed for d -21, -7, -3, 0, 1, 10, and 20 relative to calving date. To analyze absolute blood ionized Ca, electrolytes (Na^+ , K^+ , and Cl^-), and pH, blood samples on d -21, -7, -3, 0, and +1 relative to calving date were taken from the ear artery. Briefly, cows were placed in headlocker, and the backs of their ears were shaved and disinfected. Afterward, ~1 g of Tapin cream containing lidocaine (Orifarm Generics A/S) was put on the ears. After 15 min, a 21-gauge butterfly needle was used to take ~1 to 2 mL of blood from the artery into a Vacutainer that was placed on crushed ice and transferred to a laboratory located at the Danish Cattle Research Center for immediate analysis by ABL-90 Blood Gas Analyzer (Radiometer Medical ApS, Copenhagen, Denmark).

Mineral Composition Analysis

The DM content of TMR was measured by drying in an oven at 60°C for 48 h. Weekly samples of TMR were taken for mineral analysis (Vejen, Denmark; Table 1). Calcium, P, Mg, K, Na, S, Mg, Fe, Zn, and Cu in the TMR were analyzed by inductively coupled plasma MS and Cl by titration at a commercial feed testing laboratory (Eurofins Agro Testing Denmark A/S, Galten).

Plasma Metabolites and Mineral Analysis

Plasma levels of 25-OH- D_3 were determined according to Hymøller and Jensen (2011). Plasma proteins were precipitated in ethanol and methanol, and plasma triglycerides were saponified in potassium hydroxide. Then, fat-soluble vitamins were collected in heptane, and samples were dried using N_2 , and the residues were redissolved in 85% methanol. Separation and quantification of vitamin D were carried out by reverse-phase gradient HPLC and UV-detection at 265 nm, using 1 α -hydroxycholecalciferol (Sigma-Aldrich) as the internal standard. Separation and quantification were carried out by HPLC as described by Hymøller and Jensen (2011).

In plasma, total Ca was determined by the Arsenazo method (Siemens CA_2, Siemens Healthcare Diagnostics Manufacturing Ltd., Dublin, Ireland), inorganic phosphate was determined by the molybdate method (Siemens IP), total Mg was determined using the xylydyl blue method (Siemens MG), D-3-hydroxybutyrate was determined using hydroxybutyrate dehydrogenase (HBD-301; Sorachim, Lausanne, Switzerland), and nonesterified fatty acids (NEFA) were measured using the Wako NEFA C ASA-ACOD assay (Wako Chemicals GmbH, Neuss, Germany) using an ADVIA 1800 clinical chemistry analyzer (Siemens Healthcare A/S, Denmark). Plasma 1,25-(OH) $_2\text{D}_3$ was analyzed using a 1,25-(OH) $_2$ -Vitamin D ELISA assay (Immundiagnostik AG, Bensheim, Germany).

Health Records

Retained placenta, metritis, dystocia, mastitis, and milk fever were recorded according to standard procedure in Danish Cattle Research Center. For mastitis, all the cows were checked twice a day in milking parlor for any sign of mastitis. Any assistance from staff during calving was recorded as dystocia. Failure to expel fetal membranes within 24 h after parturition was recorded as retained placenta, and a veterinarian was immediately called for treatment. At ~14 d after calving, all cows were checked by a veterinarian for metritis. In the first 2 d after calving, cows with very low DMI, low body temperature, and finally lack of ability to stand up were recorded as milk fever.

Statistical Analysis

The obtained data (DMI, milk yield and composition, BW, blood and plasma metabolites) were analyzed by using proc mixed procedure in SAS (version 9.4) with maximum likelihood method, and the model was

$$Y_{ijk} = \mu + D_i + T_j + D \times T_{ij} + c_k + e_{ijk},$$

where Y is the dependent variable, μ is the overall mean, and the model includes fixed effect of D th dietary treatment (i ; D_3 or HyD), T th of time (j ; days of week from 21 d before parturition to 20 d after), interaction between dietary treatment and time, random effect of k th cow (c_k ; 21 cows per treatment), and the random error (e_{ijk}). The approximation for df was specified by DDFM = SAT-TERTHWAITE option in the stated model. Individual cows were experiment unit, and time was also handled as repeated in the model. When the interaction between treatment and time or significant effect of treatment was found, lsmeans were compared within the same days by slice option by using pairwise differences (PDIF).

For milk fever, metritis, dystocia, mastitis, and retained placenta a dataset was prepared with “yes” and “no,” and incidence of each case was compared using Fisher’s exact test in SAS by using proc Freq.

Except for Table 1, values presented in tables and figures are lsmeans with corresponding SEM. All the data were normally distributed except for SCC and NEFA, which were transformed by $\log_{10}$ and square root, respectively, and the transformed data and CI were reported. Significance was declared at $P \leq 0.05$ and a tendency of $0.05 < P \leq 0.10$. All 42 cows were included for the analysis of artery blood electrolytes and pH. However, there were 2 dropout cows postpartum due to nonrelated treatment (2 culled cows due to leg and hoof problems, 1 cow per treatment), and the 40 cows were included in the data analysis of the rest of the parameters. The BCS was

Table 2. Milk yield, milk composition, BW, and BCS of cows fed either D₃ (n = 20) or 25-OH-D₃ (n = 20) during the transition period

Item	Treatment		SEM or CI ¹	P-value		
	D ₃	HyD		Treatment	Time	Interaction
DMI, kg/d						
From wk -3 to -2 before calving	15.2	13.6	0.44	0.841	<0.001	0.0001
From wk -2 to -1 before calving	16.4	14.4				
From wk -1 to calving	14.6	13.9				
From calving to wk +1	16.1	17.1				
From wk +1 to +2	19.5	20.2				
From wk +2 to +3	21.3	22.9				
Milk yield, kg/d						
Wk 1	32.7	32.5	1.43	0.503	<0.001	0.166
Wk 2	38.3	36.8				
Wk 3	41.7	39.7				
ECM, kg/d						
Wk 1	37.1	40.2	2.12	0.976	<0.001	0.095
Wk 2	43.0	41.7				
Wk 3	44.2	43.4				
Milk fat, %						
Wk 1	5.2	5.5	0.17	0.719	<0.001	0.083
Wk 2	5.2	5.0				
Wk 3	4.8	4.6				
Milk protein, %						
Wk 1	4.5	4.3	0.08	0.236	<0.001	0.229
Wk 2	3.9	3.8				
Wk 3	3.6	3.6				
Milk lactose, %						
Wk 1	4.6	4.7	0.03	0.123	<0.001	0.682
Wk 2	4.8	4.8				
Wk 3	4.9	5.0				
SCC, ² per mL						
Wk 1	2.6	2.4	2.1–2.6	0.333	<0.001	0.891
Wk 2	2.4	2.2				
Wk 3	2.2	2.0				
Milk fat to protein ratio						
Wk 1	1.17	1.28	1.28	0.913	0.022	0.064
Wk 2	1.34	1.32				
Wk 3	1.33	1.28				
Milk BHB, mM						
Wk 1	0.023	0.049	0.01	0.699	0.006	0.045
Wk 2	0.055	0.052				
Wk 3	0.063	0.056				
Milk acetone, mM						
Wk 1	0.016	0.036	0.01	0.880	0.895	0.063
Wk 2	0.024	0.027				
Wk 3	0.036	0.022				
BW, kg						
Wk 1	722	741	12	0.167	<0.001	0.290
Wk 2	709	730				
Wk 3	699	725				
BCS (1–5 scale)	3.3	3.3	0.03	0.129	—	—

¹SEM or 95% CI for treatment.²Somatic cell counts were transformed by log₁₀. No significant difference was observed between lsmeans in the same days.

analyzed by a general linear model with treatment as the only source of variation.

RESULTS

Productive Performance and Health Records

The productive performance of D₃ and HyD is presented in Table 2. The length of pregnancy was similar

between the D₃ (279 ± 0.96) and HyD (277 ± 0.91; means ± SEM, *P* = 0.19). There was an interaction between treatment and time for DMI (*P* < 0.001; Figure 1). The DMI in the HyD group was lower than in the D₃ group at d -3, -2, and -1 relative to calving; however, the DMI in HyD group was numerically higher than the D₃ group at d 0, 1, and 2 postpartum. On d -1 relative to calving, the DMI in D₃ group was 1.3 kg higher than the HyD group. However, on d 1 postcalving, the DMI was 2.6 kg higher

in the HyD group than in the D₃ group ($P = 0.017$ for lsmeans comparison). As expected, milk yield was not affected by treatment. However, there was a tendency for interaction between treatment and time for ECM (kg/d; $P = 0.095$). The ECM was higher in HyD than in D₃ in wk 1 postpartum; however, in wk 2 and 3, an opposite trend was observed. There was a tendency for interaction between treatment and time for milk fat percentage ($P = 0.083$), milk BHB ($P = 0.045$), and milk acetone ($P = 0.063$). Milk protein percentage, milk lactose percentage, and milk SCC were not affected by treatment. Milk fat ($P < 0.001$) and protein percentage ($P < 0.001$) were affected by time and decreased from wk 1 to 3 postpartum. However, milk lactose percentage increased from wk 1 to 3 postpartum ($P < 0.001$).

Incidences of common transitional disorders are listed in Table 3. Three cases of milk fever were observed in D₃ compared with no cases in HyD. There was a tendency for lower incidences of mastitis in HyD (1 case) compared with D₃ (5 cases; $P = 0.081$). Incidences of metritis, retained placenta, and dystocia were noted but not significantly affected by treatment.

Arterial Blood Ionized Ca and Plasma Vitamin D Metabolites

Arterial blood ionized Ca was affected by treatment (Figure 2; $P = 0.045$), and it was significantly higher in the HyD (1.08) compared with D₃ (0.99 mmol/L; $P = 0.002$ for lsmeans comparison) on calving day (d 0). Plasma 25-OH-D₃ was affected by treatment ($P < 0.001$) and time ($P < 0.001$; Figure 3), and there was an interaction between treatment and time ($P < 0.001$). However, plasma 1,25(OH)₂D₃ was not affected by treatment, and it increased from d -3 to 1 relative to calving ($P < 0.001$ for time effect). The plasma 25-OH-D₃ level on d -7, -3, 0, 1, 10, and 20 relative to calving was significantly higher in the HyD than the D₃ group ($P < 0.001$ for lsmeans comparison). The plasma 25-OH-D₂ level decreased from d -21 to 20 relative to calving ($P < 0.001$ for time effect).

Blood and Plasma Metabolites

Plasma minerals and metabolites are presented in Table 4. Plasma level of total Ca ($P = 0.405$), inorganic P ($P = 0.323$), and Mg ($P = 0.988$) was not affected by treatment; however, there was a significant effect of time ($P < 0.001$ for all 3 plasma minerals) without interaction between treatment and time. Plasma levels of total Ca and inorganic P decreased from d -21 to 0 relative to calving and increased from d 1 to 20 relative to calving.

Plasma NEFA level was not affected by treatment. Plasma NEFA level was affected by time ($P < 0.001$), increasing from d -7 to 10 relative to calving. Plasma

Table 3. Health records of cows fed either D₃ (n = 20) or 25-OH-D₃ (n = 20) during the transition period

Item	Treatment		P-value
	D ₃	HyD	Fisher's exact test probability
Milk fever	3	0	0.115
Metritis	1	0	0.500
Dystocia	1	1	0.513
Mastitis	5	1	0.081
Retained placenta	2	0	0.244

BHB level was affected by treatments ($P = 0.004$), being lower in HyD (0.61) compared with D₃ (0.70 mM).

Arterial blood glucose and lactate levels were not affected by treatment (Table 5). Arterial blood glucose level was affected by time ($P < 0.001$), increasing from d -21 to 0 relative to calving. Similarly, arterial blood lactate level was affected by time ($P < 0.001$), increasing from d -21 to 1 relative to calving. Treatment did not affect the arterial blood Na⁺, Cl⁻, K⁺, extracellular base excess, bicarbonate, and pH; however, there was a significant effect of time (Table 5; $P < 0.001$ for all). Arterial blood Na⁺ and Cl⁻ increased from d -21 to 0 relative to calving and decreased on d 1 relative to calving. In contrast, arterial blood K⁺, bicarbonate, and pH decreased from d -21 to 0 relative to calving and increased on d 1 relative to calving.

DISCUSSION

Productive Performance and Health Records

Performance results in Table 2 should be interpreted with caution due to limited number of cows. The interaction between treatment and time showed that the HyD had lower intake before calving and higher DMI after calving. Poindexter et al. (2023a) reported a similar prepartum DMI in cows fed 3 mg/d of 25-OH-D₃ (8.8) compared with 3 mg/d of D₃ (9.1 kg/d) from 250 d of gestation until parturition. However, Scanga et al. (2001) reported that feed intake significantly reduced in feedlot cattle fed a high dose of 100 mg/d of D₃ compared with the unsupplemented group. Milk yield was not affected by treatment. In line with our results, Martinez et al. (2018b) reported that milk yield was not affected by supplementation of 3 mg/d of 25-OH-D₃ compared with D₃ from 252 d of gestation until calving. There was a tendency for interaction between treatment and time for milk fat percentage, and milk fat percentage was higher in HyD than in D₃ in wk 1 postpartum. In line with milk fat percentage, a similar tendency for interaction between treatment and time was observed for milk BHB. However, the plasma NEFA and BHB concentrations did not follow the same pattern as milk BHB and acetone, as

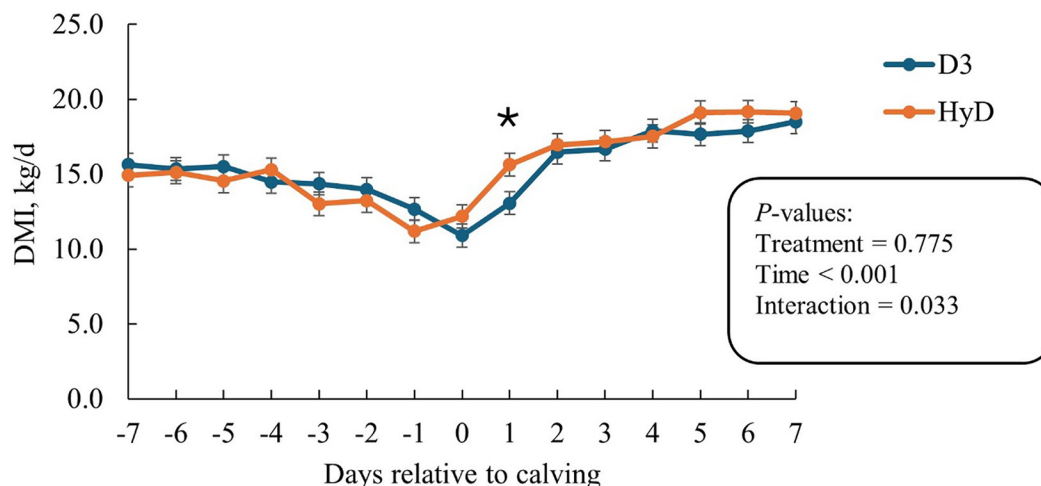


Figure 1. The DMI of cows fed either D₃ (n = 20) or 25-OH-D₃ (n = 20) during the transition period. *Shows significant difference on d 1 ($P = 0.017$ for lsmeans comparison). Error bars represent SEM. Treatment SEM was 0.5.

the plasma samples were collected on different days than the milk samples during the experiment.

In the present study, indeed, it is a rather limited dataset for evaluating risk of illnesses and metabolic disorders. However, it was interesting to note that there was a numerically lower incidence of milk fever in HyD compared with D₃. In addition, the numerically lower SCC was in agreement with tendency for lower incidence of mastitis in HyD compared with D₃. It has been suggested that 25-OH-D₃ supplementation may reduce the length of the inflammatory response by improving immune cell response and lowering detrimental effects of unnecessary inflammatory responses (Poindexter et al., 2023b). Thus, this may explain the low numerical incidence of mastitis in HyD compared with the D₃. This beneficial effect of 25-OH-D₃ supplementation on lower numerical incidence of milk fever can be explained by significantly higher arterial blood ionized Ca on calving day, resulting in reduced risk of hypocalcemia (Poindexter et al., 2023b). Similar to our results, Martinez et al. (2018a) reported a lower incidence of retained placenta and metritis in cows fed 3 mg/d of 25-OH-D₃ compared with 3 mg/d D₃ from 252 d of gestation until calving.

Arterial Blood Ionized Ca and Plasma Vitamin D Metabolites

The results of the present study showed that supplementation of 25-OH-D₃ in the diet was more effective than D₃ supplementation at increasing concentrations of plasma 25-OH-D₃. Additionally, it is important to note that part of the increased plasma 25-OH-D₃ levels in HyD compared with D₃ group could be due to higher

supplementation level of 25-OH-D₃ compared with D₃. In line with our findings, feeding either 1 or 3 mg/d of 25-OH-D₃ progressively increased plasma 25-OH-D₃ compared with 1 or 3 mg/d of D₃ in both pre- and postpartum (Vieira-Neto et al., 2021). The most important result from the current study was that HyD group, compared with D₃ group, had a significantly higher arterial blood ionized Ca level at calving day. However, arterial blood ionized Ca level prepartum did not increase, which can be explained by increased urinary Ca excretion (Rodney et al., 2018; Poindexter et al., 2023b). Wilkens et al. (2012) reported that urine Ca excretion normalized for creatinine was significantly higher in cows receiving 3 mg of 25-OH-D₃/d compared with nonsupplemented cows in both diets with high (+144) and low (−168 mEq/kg DM) DCAD. In this study, we observed an interaction between treatment and time for DMI. The HyD group had lower DMI on d −3, −2, and −1 relative to calving but higher DMI on d 0, 1, and 2 compared with D₃. This increased DMI in the HyD could partly explain the higher arterial blood ionized Ca levels on d 1 and 2 postpartum.

To the best of our knowledge, the current study is one of the first studies to analyze the effects of feeding 25-OH-D₃ based on arterial blood sampling in dairy cows. In previous studies evaluating dietary supplementation effect of 25-OH-D₃ (Wilkens et al., 2012; Silva et al., 2022), ionized Ca was measured from coccygeal venous blood, which differs from arterial blood. This difference is important because the regulation of ionized Ca via parathyroid hormone and 1,25(OH)₂D₃ via kidney is driven by the concentration of ionized Ca in arterial blood. Moreover, differences in pH between arterial and venous blood can influence ionized Ca levels.

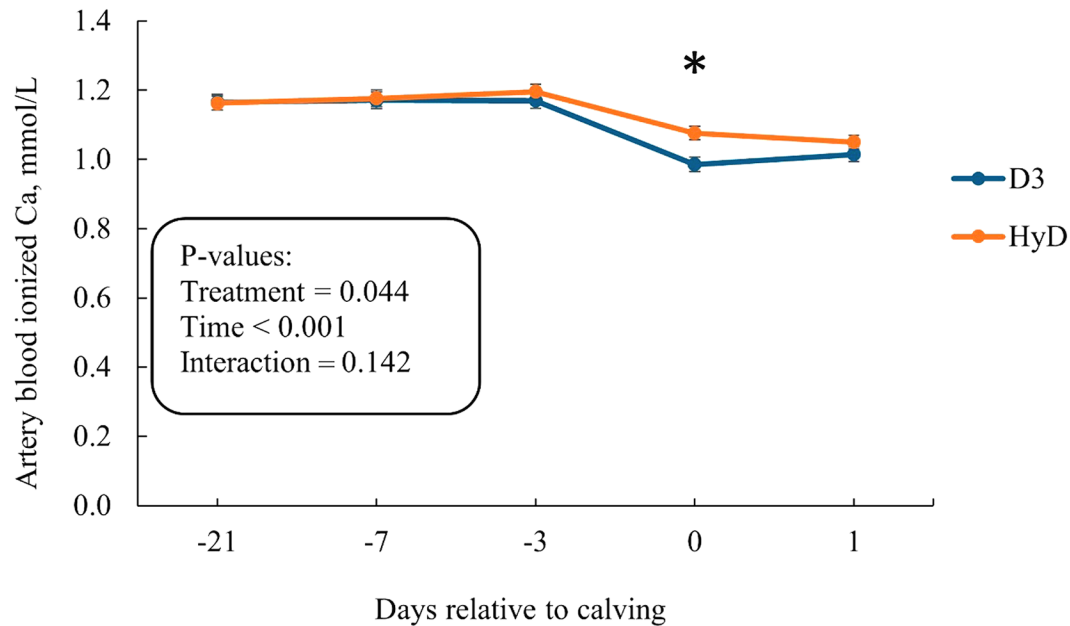


Figure 2. Artery blood ionized Ca of cows fed either D₃ (n = 21) or 25-OH-D₃ (n = 21) during the transition period. *Shows significant difference on d 0 (13 ± 8 h after calving; *P* = 0.002 for lsmeans comparison). Error bars represent SEM. Treatment SEM was 0.01.

Therefore, interpreting vitamin D effect is more precise when based on arterial ionized Ca, as this reflects the Ca concentration entering the kidneys for 1,25(OH)₂D₃ synthesis and the parathyroid glands for parathyroid hormone secretion.

On calving day, the HyD had higher arterial blood ionized Ca and plasma 25-OH-D₃ levels compared with the D₃. This effect seemed to be the direct effect of 25-OH-D₃ on Ca metabolism. An earlier in vitro study has shown that perfusion of 2.5 µg of 25-OH-D₃ added to the intestines of vitamin D-deficient rats via the arterial blood perfusate increased Ca transport levels within 2 h (Olson and DeLuca, 1969). In contrast, 250 µg of cholecalciferol had no effect on Ca transport levels over a 4-h period. Olson and DeLuca (1969) concluded that there is evidence for 25-OH-D₃ being a metabolically active form of vitamin D₃. In fact, the conclusion by Olson and DeLuca (1969) was supported by other studies (Brickman et al., 1976; Colodro et al., 1978). These studies measured ⁴⁷Ca absorption in response to different doses of 25-OH-D₃ and 1,25(OH)₂D₃ and found that under normal conditions, the molar potency of 25-OH-D₃ was only 2 orders of magnitude lower than that of 1,25(OH)₂D₃. This relative potency demonstrated (Brickman et al., 1976; Colodro et al., 1978; Heaney et al., 1997) that 25-OH-D₃ might actively contribute to vitamin D-like activity, which can be an explanation for increased arterial blood ionized Ca level on calving day. Additionally, the metabolic role of other metabolites of vitamin D should be considered (Combs and McClung,

2016). As an example, van Leeuwen et al. (2001) reported that the 24-hydroxylated vitamin D metabolites, such as 24,25(OH)₂D₃, are not merely a degradation product but also have distinct effects on bone. This suggests that metabolites such as 24,25(OH)₂D₃ may contribute to the physiological effects observed at high doses of 25-OH-D₃.

The increased arterial blood ionized Ca level in the D₃ after calving can be explained by the indirect effect of 25-OH-D₃ on bone Ca resorption. Rodney et al. (2018) reported that blood serotonin was increased from 1,543 to 1,714 (µg/mL) in cows fed 3 mg/d of D₃ or 25-OH-D₃ in a diet with DCAD of -130 (mEq/kg), which plays a role in bone remodeling and Ca transport during lactation (Chabbi-Achengli et al., 2012; Laporta et al., 2014a). Serotonin triggers parathyroid-related protein production, stimulating osteoclast activity and bone mobilization (Laporta et al., 2014b). Wilkens et al. (2014) noted that a lower carboxylated osteocalcin to C-terminal telopeptide of type 1 collagen ratio indicates increased bone resorption. Rodney et al. (2018) reported a higher postpartum ratio in cows fed 3 mg/d of 25-OH-D₃ compared with those fed D₃.

It has been reported that dietary supplementation of 25-OH-D₃ during the peripartum period can bypass renal 1α-hydroxylase feedback, significantly increasing plasma 1,25-(OH)₂-D₃ postpartum (Wilkens et al., 2012; Weiss et al., 2015) to meet the high Ca demand for colostrum and milk synthesis. However, in the present study, the increase in plasma 25-OH-D₃ did not lead

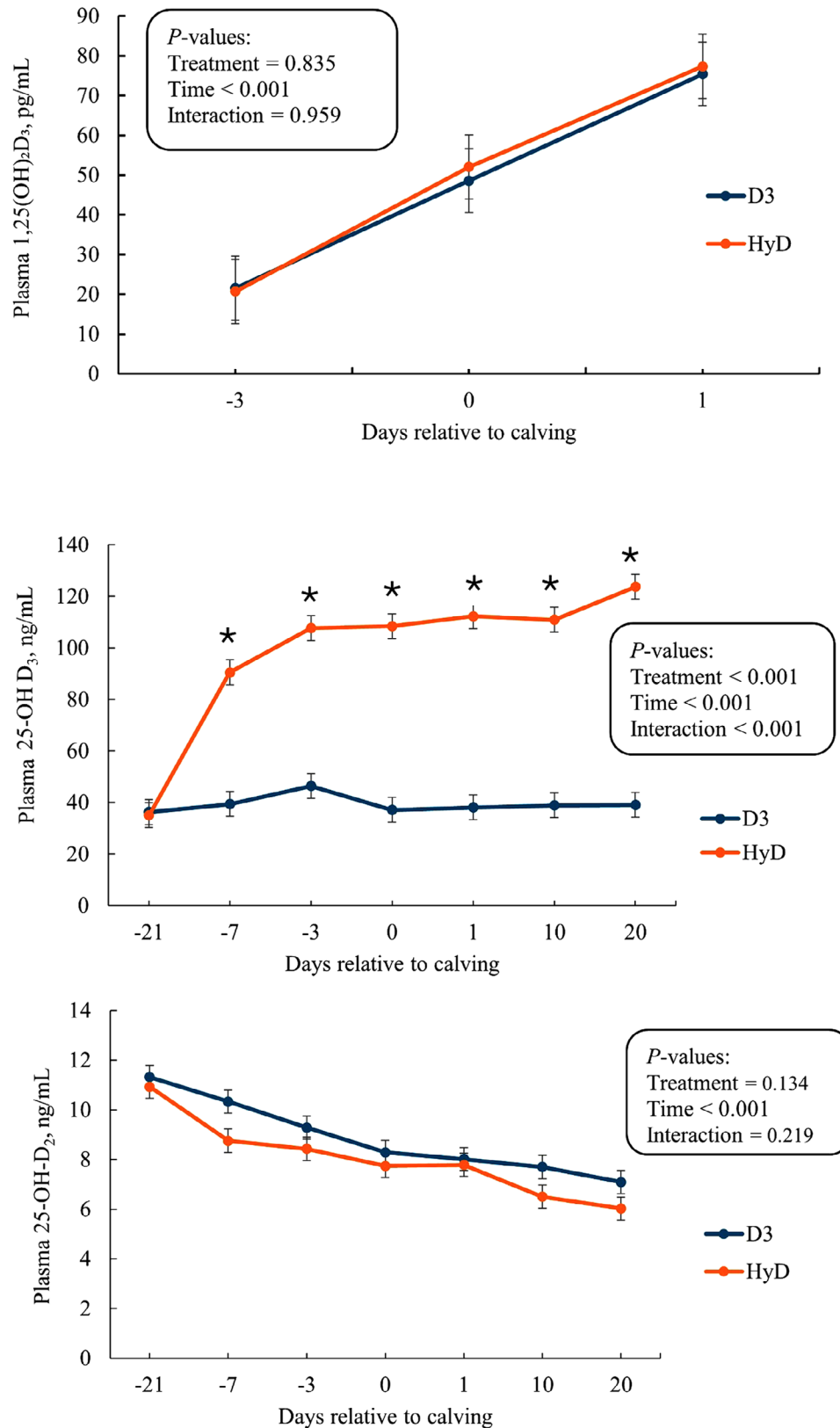


Figure 3. Plasma 1,25(OH)₂D₃ (upper panel), 25-OH-D₃ (middle panel), and 25-OH-D₂ (lower panel) of cows fed either D₃ (n = 20) or 25-OH-D₃ (n = 20) during the transition period. *Shows significant difference ($P < 0.001$ for all lsmeans comparison). Error bars represent SEM. Treatment SEM was 4.2, 4.1, and 0.39 for plasma 1,25(OH)₂D₃, 25-OH-D₃, and 25-OH-D₂, respectively.

Table 4. Plasma mineral and metabolites of cows fed either D₃ (n = 20) or 25-OH-D₃ (n = 20) during the transition period

Item	Treatment		SEM or CI ¹	P-value		
	D ₃	HyD		Treatment	Time	Interaction
Total Ca, mM						
21 d before calving	2.31	2.27	0.09	0.405	<0.001	0.131
7 d before calving	2.18	2.27				
3 d before calving	2.05	2.30				
Calving day	1.92	1.95				
1 d after calving	2.17	2.27				
10 d after calving	2.17	2.46				
20 d after calving	2.31	2.27				
Inorganic P, mM						
21 d before calving	1.73	1.67	0.05	0.323	<0.001	0.167
7 d before calving	1.83	1.80				
3 d before calving	1.78	1.91				
Calving day	1.20	1.36				
1 d after calving	1.46	1.39				
10 d after calving	1.44	1.62				
20 d after calving	1.73	1.67				
Mg, mM						
21 d before calving	0.83	0.78	0.03	0.988	<0.001	0.045
7 d before calving	0.80	0.84				
3 d before calving	0.79	0.81				
Calving day	0.86	0.82				
1 d after calving	0.76	0.77				
10 d after calving	0.81	0.87				
20 d after calving	0.83	0.78				
NEFA, ² mEq/L						
21 d before calving	6.8	7.4	13.1–16.5	0.801	<0.001	0.335
7 d before calving	4.9	5.9				
3 d before calving	8.7	8.6				
Calving day	19.8	16.3				
1 d after calving	21.1	23.4				
10 d after calving	19.7	18.7				
20 d after calving	6.8	7.4				
BHB, mM						
21 d before calving	0.54	0.45	0.02	0.004	<0.001	0.448
7 d before calving	0.78	0.61				
3 d before calving	0.69	0.55				
Calving day	0.69	0.67				
1 d after calving	0.80	0.70				
10 d after calving	0.70	0.60				
20 d after calving	0.54	0.45				

¹SEM or 95% CI for treatment.²NEFA data were transformed by square root. No significant difference was observed between lsmeans in the same days.

to a corresponding increase in plasma 1,25(OH)₂D₃. Differences in the dietary dose of 25-OH-D₃ and DCAD in the close-up diet in the current study, compared with the dose of 25-OH-D₃ (3 mg/d) and DCAD (–168 mEq/kg DM) in Wilkens et al. (2012) and the dose of 25-OH-D₃ (6 mg/d) and DCAD (–139 mEq/kg DM) in Weiss et al. (2015), may explain the variation in serum/plasma 1,25-(OH)₂D₃ responses. In contrast to our findings, Rodney et al. (2018) showed that high doses of 25-OH-D₃ in dairy cow diets can increase plasma 1,25(OH)₂D₃. The reason for the lack of effect of 25-OH-D₃ on 1,25(OH)₂D₃ could be due to the tight physiological regulation of the enzyme 1 α -hydroxylase, which converts 25-OH-D₃ to its active form, 1,25(OH)₂D₃ (Fraser and

Kodicek, 1973), and maintenance of normal levels of total Ca and inorganic P. During the entire experiment, except for calving day, the plasma levels of total Ca and inorganic P were within the normal range of 2.2 to 2.5 mM and 1.3 to 2.6 mM, respectively (NASEM, 2021). Bouillon et al. (2022) reported when Ca and phosphate levels are adequate, or parathyroid hormone is low, the activity of 1 α -hydroxylase is suppressed, thereby limiting the conversion of 25-OH-D₃ to 1,25(OH)₂D₃, even in the presence of sufficient 25-OH-D₃. In the HyD group, on the day of calving, plasma levels of 25-OH-D₃ were ~1,500 times higher than those of 1,25(OH)₂D₃, suggesting that 25-OH-D₃ may not be a limiting factor for the synthesis of 1,25(OH)₂D₃.

Table 5. Artery blood metabolites, electrolytes, and pH of cows fed either D₃ (n = 21) or 25-OH-D₃ (n = 21) during the transition period

Item	Treatment			P-value		
	D ₃	HyD	SEM ¹	Treatment	Time	Interaction
Artery blood glucose, mM						
21 d before calving	3.9	3.9	0.13	0.728	<0.001	0.493
7 d before calving	4.0	4.2				
3 d before calving	4.2	4.2				
Calving day	5.5	4.9				
1 d after calving	4.0	4.1				
Artery blood lactate, mM						
21 d before calving	0.60	0.41	0.07	0.113	<0.001	0.602
7 d before calving	0.44	0.42				
3 d before calving	0.47	0.51				
Calving day	0.93	0.76				
1 d after calving	0.87	0.48				
Na ⁺ , mM						
21 d before calving	142	142	0.2	0.177	<0.001	0.658
7 d before calving	146	145				
3 d before calving	147	146				
Calving day	148	147				
1 d after calving	146	147				
Cl ⁻ , mM						
21 d before calving	102	101	0.4	0.770	<0.001	0.933
7 d before calving	107	107				
3 d before calving	108	108				
Calving day	108	107				
1 d after calving	102	102				
K ⁺ , mM						
21 d before calving	4.1	4.1	0.03	0.287	0.002	0.423
7 d before calving	4.0	4.0				
3 d before calving	4.0	4.0				
Calving day	4.0	4.0				
1 d after calving	4.2	4.1				
Extracellular base excess, mM						
21 d before calving	4.2	5.4	0.44	0.410	<0.001	0.48
7 d before calving	1.1	0.5				
3 d before calving	2.5	0.9				
Calving day	3.7	2.9				
1 d after calving	9.4	8.6				
Bicarbonate (HCO ₃ ⁻), mM						
21 d before calving	28.1	29.1	0.33	0.641	<0.001	0.388
7 d before calving	25.5	25.2				
3 d before calving	26.7	25.4				
Calving day	27.7	27.3				
1 d after calving	31.8	31.8				
pH						
21 d before calving	7.48	7.50	0.004	0.877	<0.001	0.365
7 d before calving	7.45	7.46				
3 d before calving	7.47	7.45				
Calving day	7.48	7.47				
1 d after calving	7.50	7.51				

¹SEM for treatment. No significant difference was observed between lsmeans in the same days.

Blood and Plasma Metabolites

Despite the effect of HyD on arterial blood ionized Ca, total plasma Ca levels were unaffected by treatment. Arterial blood ionized Ca represents metabolically active Ca, so conclusions based on plasma total Ca cannot directly reflect functional Ca status. Blood pH, a key factor affecting ionized Ca levels (Wang et al., 2002), was not a driving factor here, as arterial blood pH and

electrolytes were similar in D₃ and HyD. The lack of a parallel increase in arterial blood ionized Ca and plasma total Ca on d 0 and the first day postcalving in the HyD group suggests plasma total Ca may have reached its upper threshold, limiting further increases.

The normal range of plasma inorganic P (Anderson et al., 2017) and Mg (NASEM, 2021) in cows is 1.3 to 2.6 and 1.1 to 2.3 mM, respectively. Blood levels of these minerals respond directly to dietary P (Horst, 1986) and

Mg (Jittakhot et al., 2004) intake. In the present study, the lack of significant treatment effect on plasma inorganic P and Mg could be explained by the similar P and Mg levels in the diets.

In agreement with the results of the present study, Silva et al. (2022) reported a significantly lower plasma NEFA and numerically lower BHB in cows fed 3 mg/d of 25-OH-D₃ compared with 0.625 mg/d of D₃ from 30 d before expected calving day to 42 DIM in diets with DCAD of −110 mEq/kg. There was a lower numerical BW loss (16 kg) in HyD compared with D₃ (23 kg) from wk 1 to 3 postpartum, which was in line with numerically lower plasma NEFA, except for d 0, and significantly lower plasma BHB level. This study found no effect of treatment on arterial blood glucose levels, consistent with Poindexter et al. (2020), who reported similar serum glucose levels in cows supplemented with 1 or 3 mg of 25-OH-D₃ compared with D₃ prepartum. However, Silva et al. (2022) found higher plasma glucose levels in cows fed 3 mg/d of 25-OH-D₃ compared with 0.625 mg/d of D₃.

Changes in blood electrolytes and pH over time are likely due to alterations in DCAD at pre- and postpartum. In this study, cows fed moderate negative DCAD diets prepartum had reduced blood levels of Na⁺, K⁺, extracellular base excess, HCO₃[−], and pH. Postpartum, these levels increased due to the shift to positive DCAD. Rodney et al. (2018) studied the effect of high (+130) and low (−130 mEq/kg DM) DCAD supplemented with 3 mg/d of either D₃ or 25-OH-D₃ and clearly showed that DCAD was the main driver for changes in blood electrolytes and pH.

CONCLUSIONS

Supplementing transition dairy cows with HyD raised plasma 25-OH-D₃ compared with cows supplemented with D₃ without affecting plasma 1,25(OH)₂D₃ concentrations. An interaction between treatment and time was observed for DMI showing reduced intake immediately prepartum and increased intake immediately postpartum with HyD. Blood ionized Ca concentrations were higher at calving day for HyD.

NOTES

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Nonstandard abbreviations used: D₃ = cholecalciferol; HyD = 25-OH-D₃; NEFA = nonesterified fatty acids.

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